

both nulls at conventional significance levels (5%). We also reject both hypotheses in the post-programme period for all demographic groups except for young males.²¹

Note that tests of mean independence of the residuals conducted on pre-programme earnings do not always produce the same inference as tests conducted on post-programme earnings. See especially the results for adult males or young males in Table 3(a). It is, therefore, not a safe strategy to use pre-programme tests about mean selection bias to make inferences about post-programme selection bias, as proposed by Heckman and Hotz (1989).

Table 3(b) presents test results for the weaker hypothesis of conditional mean independence in the pre- and post-programme differences, hypotheses ($H_{(A-S)}$) and the hypothesis $H_{(A-S')}$. These hypotheses are formulated in terms of symmetric differences around the date of enrollment into the programme, $t=0$, so $t=-t'$ where t' is a pre-programme period and t is a post-programme period. We do not reject these null hypotheses for any demographic group suggesting that the semiparametric conditional method of difference-in-differences proposed in this paper is consistent with the data.

In summary, we test and reject the conditional independence assumptions (A-3) maintained in the literature and the weaker mean independence assumption (A-4). The hypothesis of mean independence of U_0 is somewhat more concordant with the data than mean independence of the raw data (Y_0) but it is still rejected in most instances. Finally we test, and do not reject, the identifying assumptions for the conditional difference-in-differences estimator for symmetric differences around the date of enrollment into the programme.

11. ALTERNATIVE MATCHING ESTIMATORS

All matching estimators, including our semiparametric conditional difference-in-differences estimator, can be cast in the following framework

$$\hat{M}(S) = \sum_{i \in I_1} \omega_{N_0, N_1}(i) [Q_{1i} - \sum_{j \in I_0} W_{N_0, N_1}(i, j) Q_{0j}], \quad \text{for } X \in S, \quad (10)$$

where Q_{1i} is a treatment outcome and Q_{0j} is a comparison group outcome. Q_{1i} and Q_{0j} sometimes represent Y_{1i} and Y_{0j} , respectively, but the notation is more general to allow us to use regression-adjusted Y_{1i} and Y_{0j} . N_1 is the number of programme participants, N_0 is the number of persons in the comparison group, $W_{N_0, N_1}(i, j)$ is a weight with $\sum_{j \in I_0} W_{N_0, N_1}(i, j) = 1$, and $\omega_{N_0, N_1}(i)$ is a weight that accounts for heteroscedasticity and scale. I_1 is the set of indices for programme participants and I_0 a set of indices for comparison group members. Matches for each participant are constructed by taking weighted averages over comparison group members. Matching can be performed within various strata to recover estimates for different populations of interest.

Matching estimators differ in the weights they attach to members of the comparison group. Define a neighbourhood $C(X_i)$ for person i in the participant sample, $i \in I_1$. Neighbours for i are persons $j \in I_0$ for whom $X_j \in C(X_i)$. Persons matched to i are those people in set A_i where $A_i = \{j \in I_0 \mid X_j \in C(X_i)\}$.

A nearest-neighbour matching estimator sets $Q_{1i} = Y_{1i}$, $Q_{0j} = Y_{0j}$, $\omega_{N_0, N_1}(i) = 1/N_1$ and for each i in the $D=1$ sample picks the match $C(X_i) = \min_j \|X_i - X_j\|$, $j \in I_0$ where $\|\cdot\|$ is

21. We raise one cautionary methodological note. The asymptotic theory suggests that estimation of the parameters of P and of the parameters of the outcome equation (β) is irrelevant for construction of the test statistics. This phenomenon arises because the parameters of these functions converge at a faster rate than the nonparametric functions that we use in the tests. Yet in a small scale Monte Carlo study we find that inferences from the residuals of fitted models are sensitive to parameter estimation. This problem does not affect the tests based on unadjusted earnings, only those based on residuals. See the discussion in Appendix B.

a norm. A_i is a singleton set, except for ties that are broken by a random draw. The weighting scheme for the nearest neighbour estimator assigns all the weight to the single match: $W_{N_0, N_1}(i, j) = 1$ if $j \in A_i$; $W_{N_0, N_1}(i, j) = 0$ otherwise. Two versions of this method are (a) X_j may be reused for other matches (sampling with replacement); (b) X_j may not be reused (sampling without replacement).

The distance between persons i and j can be substantial if $C(X_i)$ is not restricted. Caliper matching avoids the problem of a substantial gap between X_i and X_j . (Cochrane and Rubin (1973).) Matches are made to i only if

$$\|X_i - X_j\| < \varepsilon, \quad j \in I_0,$$

where ε is a pre-specified tolerance. Otherwise no match is made and person i is left unmatched. The neighbourhood is $C(X_i) = \{X_j | \|X_i - X_j\| < \varepsilon\}$. If more than one person is in A_i , then the nearest neighbour under norm $\|\cdot\|$ is selected. A variant of caliper matching selects one metric to caliper match and another to pick among elements in the control sample if more than one observation qualifies under the caliper criterion. The methods can be applied with or without reuse of the observations in the comparison group. The Mahalanobis metric is commonly used in the matching literature: $\|X_i - X_j\| = (X_i - X_j)' \sum^{-1} (X_i - X_j)$ where $\sum$ is the covariance matrix formed from the $D=1$ sample.

The kernel-based matching estimators that we use and whose asymptotic distribution is derived in our companion paper (Heckman, Ichimura and Todd (1993 revised, 1997)), construct matches by forming weighted averages of the outcomes of all individuals in the $D=0$ comparison sample. If weights from a symmetric, nonnegative, unimodal kernel are used, then the average places higher weight on persons close in terms of X_i and lower weight on more distant observations. Kernel matching sets $A_i = I_0$ and defines

$$W_{N_0, N_1}(i, j) = \frac{G_{ij}}{\sum_{k \in I_0} G_{ik}},$$

where $G_{ik} = G((X_i - X_k)/a_{N_0})$ is a kernel function and a_{N_0} is a bandwidth parameter.²² Kernel matching is a local averaging method that reuses and weights all the comparison group observations in the treatment sample.

Local linear matching uses the weight

$$W_{N_0, N_1}(i, j) = \frac{G_{ij} \sum_{k \in I_0} G_{ik} (X_k - X_i)^2 - [G_{ij} (X_j - X_i)] [\sum_{k \in I_0} G_{ik} (X_k - X_i)]}{\sum_{j \in I_0} G_{ij} \sum_{k \in I_0} G_{ik} (X_k - X_i)^2 - (\sum_{k \in I_0} G_{ik} (X_k - X_i))^2}. \quad (11)$$

We use local linear weights instead of more conventional kernel weights because local linear estimators converge at a faster rate at boundary points and adapt better to different data densities. A substantial fraction of our data is at boundaries. See Figure 1.²³

The regression-adjusted local linear matching estimator developed in Heckman, Ichimura, Smith and Todd (1994, revised 1996b) and in Heckman, Ichimura and Todd (1993, revised 1997) combines local linear matching on the probability of participation $P(X)$ and regression adjustment on the X . It extends classical matching methods by utilizing information on the functional form of outcome equations and by incorporating exclusion

22. a_{N_0} satisfies $\lim_{N_0 \rightarrow \infty} a_{N_0} \rightarrow 0$. Precise conditions on the rate of convergence needed for consistency and asymptotic normality of the kernel matching estimator are presented in our companion paper. (Heckman, Ichimura and Todd (1993, 1997).)

23. These properties of the local linear estimator that make it superior to the standard kernel regression estimator are discussed in Fan (1992).

restrictions across outcome and participation equations. Previously-developed matching methods do not use these sources of information.

Regression-adjusted matching is performed by the following procedure. Assume a conventional econometric model for outcomes in the no-treatment state that is additively separable in observables and unobservables: $Y_0 = X\beta_0 + U_0$. Using partially linear regression methods applied to the comparison group sample, estimate the components of $E(Y_0|X, D=0) = X\beta_0 + E(U_0|X, D=0)$ imposing any desired exclusion restrictions.²⁴ To estimate $M(S)$ remove $X\hat{\beta}_0$ from both Y_0 and Y_1 . In the framework of equation (10), $Q_{1i} = (Y_{1i} - X_{i0}\hat{\beta}_0)$, $Q_{0j} = (Y_{0j} - X_{j0}\hat{\beta}_0)$ and local linear regression weight (11) is used. Our companion paper (Theorem 2), presents a proof of consistency and asymptotic normality of the matching estimator under Assumption (A-4'). This estimator is a compromise between a fully nonparametric approach, which is not likely to yield reliable estimates in samples at our disposal, and a more parametric model.²⁵

The conditional difference-in-difference matching estimator introduced in Section 4 sets $Q_{1i} = (Y_{1it} - Y_{0it'})$ and $Q_{0j} = (Y_{0jt} - Y_{0jt'})$ in equation (10) or equivalently uses

$$\hat{D}_{t,t'}(S) = \hat{M}_t(S) - \hat{M}_{t'}(S),$$

where $M_t(S)$ and $M_{t'}(S)$ are equal to $M(S)$ defined in (3), except that Y_{1j} , Y_{0j} , I_1 and I_0 now are subscripted by t or t' . A conditional difference-in-differences matching estimator that performs semi-parametric regression-adjustment on X is defined over set S for $Q_{1i} = [(Y_{1it} - X_{it}\hat{\beta}_{0t}) - (Y_{0it'} - X_{it'}\hat{\beta}_{0t'})]$ and $Q_{0j} = [(Y_{0jt} - X_{jt}\hat{\beta}_{0t}) - (Y_{0jt'} - X_{jt'}\hat{\beta}_{0t'})]$ or equivalently

$$\tilde{D}_{t,t'}(S) = \tilde{M}_t(S) - \tilde{M}_{t'}(S).$$

Regression-adjusted conditional difference-in-differences matching is an effective method in reducing bias in our data. However, it is more demanding in terms of its data requirements than the cross-sectional matching estimators because it requires pre-programme data.

Finally, note that randomization is a special case of the matching estimator (10) for which the comparison group is an experimental control group, $Q_{1i} = Y_{1i}$ and $Q_{0j} = Y_{0j}$, $\omega_{N_0, N_1}(i) = 1/N_1$, $\omega_{N_0, N_1}(i, j) = 1/N_0$ and the entire sample of controls is used as a comparison group for each experimental observation.

12. EVALUATING THE PERFORMANCE OF DIFFERENT MATCHING ESTIMATORS

We now consider the performance of widely-used statistical matching methods for estimating parameter $M(S)$. We contrast the performance of conventional methods with those of the new methods presented here and find that our new procedures—especially the conditional differences-in-differences method—are generally more effective. Nonetheless, matching is no panacea and considerable bias relative to the size of the experimentally-determined treatment effect is found for all methods. The evidence is generally more favourable for the conditional differences-in-differences estimator.

24. Estimation of the partially linear model, when local linear regression methods are used to estimate $E(U_0|X, D=0)$, is discussed in Heckman, Ichimura, Smith and Todd (1994, revised 1996b, c).

25. Rubin (1979) presents a regression-matching procedure in which regressions are first fit on participant and control samples and then residuals from the regression are matched. He does not formally justify his estimator and it is easy to demonstrate that it is inconsistent for $E(\Delta|X, D=1)$ unless $E(U_0|X, D=0)$ is linear in X , and does not depend on D . We estimate parametric β and the nonparametric component $E(U_0|X, D=0)$ jointly using the partially linear model. See Heckman, Ichimura, Smith and Todd (1996b, c).

The following matching procedures are evaluated in this paper.

1. *Simple P Nearest Neighbour Matching*: Match using the n neighbours with the closest values of P . $n=1$ defines nearest-neighbour matching which is the conventional estimator. In tables available on request, we also consider versions of nearest-neighbour estimators that average the outcomes of the nearest 5 or 10 neighbours.
2. *Local Linear P Score Matching*: Form a weighted average over the outcomes of comparison group members using local linear regression weights. A bandwidth equal to 0.06 is used in estimation of this model and all models presented in this paper.²⁶ The biweight kernel we use throughout this paper is defined in Appendix A.
3. *Regression-Adjusted Local Linear Matching*: Using X -adjusted outcomes for participants, $Y_{1t} - X_{1t}\hat{\beta}_{0t}$, we match to corresponding adjusted outcomes for nonparticipants, $Y_{0t} - X_{0t}\hat{\beta}_{0t}$ using local linear regression weights.
4. *Conditional Difference-in-Differences Matching*: Using either unadjusted or X -adjusted outcomes for participants, $Y_{1t} - X_{1t}\hat{\beta}_{0t}$, we match to corresponding unadjusted or adjusted outcomes for nonparticipants, $Y_{0t} - X_{0t}\hat{\beta}_{0t}$, at a post-programme time period t and at a pre-programme time period t' . Local linear regression weights are used to construct the matched outcomes.

In results available on request, we also investigate both smoothed and raw Mahalanobis metric estimators. They are comparable in performance to the other matching estimators and for the sake of brevity we delete discussion of them. To learn about the bias arising from the use of the different methods, we compare the outcomes of members of an experimental control group ($D=1$, $R=0$) to the matched outcomes from several candidate comparison groups ($D=0$). Since neither group receives treatment, the discrepancy between the mean control outcomes and the mean of the matched outcomes is a measure of bias for the particular estimator being investigated.

12.1. The support problem

Several sources of bias plague nonexperimental estimators: (a) selection on unobservables, (b) failure of a common support condition and (c) failure to weight treatment and comparison group members comparably. Matching estimators are not subject to bias (c) because they effectively reweight comparison group data to equate the distribution of observables in the $D=1$ and $D=0$ samples. Matching estimators may be subject to bias due to (a) or (b). Source (b) can be eliminated if matching is performed only over regions of common support. However, this requires that the parameter of interest $M(S)$ be redefined as the mean impact over the common support region, $M(S_{10})$. Thus, two different parameters are implicitly defined when experimental and nonexperimental analyses are used to evaluate the same programme: $M(S_E)$ and $M(S_{10})$, where S_E is the support in the experimental data and S_{10} is the region of common support or overlap between comparison groups and participant or control groups. A nonexperimental estimator of $M(S_{10})$ does not estimate the same parameter as is estimated by experimental methods unless $S_E = S_{10}$. In our study, the density and empirical support for P are very different for participants ($D=1$) and for

26. For symmetric, nonnegative, unimodal kernels, observations with closer P scores receive higher weight. Other bandwidths were also tried: 0.04, 0.08 and an "optimal" plug-in bandwidth and produced essentially the same results as the value for 0.06 for all groups. The 0.06 is the average optimal bandwidth which produces smoother fits than the "optimal" bandwidth. The "optimal" plug-in bandwidth is derived in our companion paper (Heckman, Ichimura and Todd (1993, 1997)).